PHYS 143 – Mechanics and Thermal Physics

Thermal Properties of Matter

How do pressure, volume and temperature describe a gas, what happens at a change of phase, and what do molecules do?

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By the end of this week, you should be able to:

1. Use the ideal-gas equation and the gas laws to calculate and compare the states of a gas.
2. Read pV and phase diagrams, and calculate the heat in a phase change (latent heat), including calorimetry.
3. State the assumptions of the kinetic theory and explain pressure and temperature in terms of molecular motion.
4. Calculate the average kinetic energy and the rms speed of molecules, and select the right model in an unfamiliar problem.

The detailed outcomes are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 18 (§18.1, §18.3, §18.5, §18.6) and §17.5.

Practice: check → example → independent → application

Two contacts (110 minutes each)**:**

A (Tuesday) DESCRIBE, then REPRESENT
How is the state of a gas described, and what do the pV and phase diagrams show?

B (Wednesday) ANALYSE, then EXPLAIN
What does heat do at a change of phase, and what do molecules do to give pressure and temperature?

Labels: **CORE** you should be able to do this; **PRACTICE** use it to develop the skill; **REFERENCE** keep it for later use.

Where we start: a five-minute check

Not marked. Answer individually, then we review.

1. Convert $25\,^\circ\mathrm{C}$ to kelvin and $2.0\,\mathrm{atm}$ to pascals ($1\,\mathrm{atm} = 1.013 \times 10^5\,\mathrm{Pa}$).
2. A force of $50\,\mathrm{N}$ acts on an area of $0.020\,\mathrm{m}^2$. Find the pressure.
3. A ball of mass $0.10\,\mathrm{kg}$ hits a wall at $20\,\mathrm{m/s}$ and bounces back at the same speed. Find the change in its momentum.
4. Find the kinetic energy of a $2.0\,\mathrm{kg}$ mass moving at $3.0\,\mathrm{m/s}$.
5. If p is proportional to $1/V$, what happens to p when V is halved?
6. How much heat raises $0.50\,\mathrm{kg}$ of water by $20\,\mathrm{K}$ ($c = 4186\,\mathrm{J/(kg\,K)}$)?

Check: brief review

PRACTICE

1. Convert $25°\mathrm{C}$ and $2.0\,\mathrm{atm}$: $298.15\,\mathrm{K}$; $2.0 \times 1.013 \times 10^5 = 2.0 \times 10^5\,\mathrm{Pa}$.
2. $50\,\mathrm{N}$ on $0.020\,\mathrm{m}^2$: $p = F/A = 50/0.020 = 2.5 \times 10^3\,\mathrm{Pa}$.
3. $0.10\,\mathrm{kg}$ ball, $20\,\mathrm{m/s}$, bounces back: $\Delta P = (-2.0) - (+2.0) = -4.0\,\mathrm{kg\,m/s}$ (magnitude $2mv$).
4. $2.0\,\mathrm{kg}$ at $3.0\,\mathrm{m/s}$: $K = \frac{1}{2}(2.0)(3.0)^2 = 9.0\,\mathrm{J}$.
5. $p \propto 1/V$, V halved: p doubles.
6. $0.50\,\mathrm{kg}$ of water, $20\,\mathrm{K}$ rise: $Q = mc\Delta T = (0.50)(4186)(20) = 4.2 \times 10^4\,\mathrm{J}$.

Kelvin and $Q = mc\Delta T$ (Lecture 9), pressure and momentum change (Lectures 3 and 7) and kinetic energy (Lectures 4 and 5) are reused. From now on momentum is written P, so that p is free for the pressure.

But what if... Lecture 9 ended by asking what happens when a body reaches its melting or boiling point (Contact B answers it)? Before that we must be able to say what *state* a gas is in, and to picture it: Contact A.

Contact A: Describe, then Represent

How is the state of a gas described, and
what do the pV and phase diagrams show?

Poll

You close the nozzle of a bicycle pump with a finger and push the handle in quickly. The air inside becomes harder to compress, and the barrel becomes warm. **Which of the quantities pressure, volume, temperature and amount of air changed?**

Predict first

Sketch the barrel before and after. Which quantities change, and which stays fixed? Does the pressure rise by the same factor as the volume falls?

p, V and T all change; the amount of air (the number of molecules) stays fixed because the nozzle is closed. One equation should connect them. We will build it from experiment, and Example 2 returns to this pump.

But what if... we can only say "the pressure rose" and "the air warmed"? To connect p, V and T we need exact words first: the **vocabulary of a gas**.

Defining the vocabulary of a gas

CORE

Quantity	What it is	What it measures	SI unit	Representation
Pressure p	the magnitude of the force per unit area that the gas exerts on a wall of its container	how hard the gas pushes on the wall	Pa $= \mathrm{N/m^2}$	an arrow perpendicular to the wall
Amount of substance n	the number of moles; one mole contains $N_A = 6.022 \times 10^{23}$ molecules	how many molecules, not how much mass	mol	a count of molecules, $N = nN_A$
Molar mass M	the mass per mole of the substance, so that $m_{\text{total}} = nM$	how heavy one mole is	kg/mol	$M = N_A m$, where m is the mass of one molecule

Interpret: pressure answers *how hard does the gas push on its container?* Moles answer *how many molecules?*; the molar mass *how heavy is one mole?* A gauge reads $p - p_{\text{atm}}$; the gas laws use the absolute pressure p.

But what if... a sample is weighed in kilograms, but we need molecules? We must link **mass, moles and molecules**.

Rule: amount of substance

$$m_{\text{total}} = nM \qquad M = N_A m \qquad N = nN_A$$

$N_A = 6.022 \times 10^{23}\,\text{mol}^{-1}$ (Avogadro's number, the number of molecules in a mole); M in kg/mol; m is the mass of one molecule, so $m = M/N_A$.

Examples of molar mass:
nitrogen N_2: $28.0\,\text{g/mol}$; carbon dioxide CO_2: $44.0\,\text{g/mol}$; helium (monatomic): $4.00\,\text{g/mol}$.

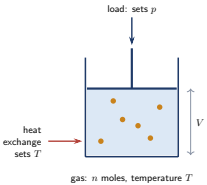
$$m(N_2) = \frac{28.0 \times 10^{-3}}{6.022 \times 10^{23}} = 4.65 \times 10^{-26}\,\text{kg};$$
$$m(CO_2) = 7.31 \times 10^{-26}\,\text{kg}.$$

Common error

Equal *masses* of two gases do not contain equal numbers of molecules. Equal *moles* do. $10\,\text{g}$ of hydrogen has many more molecules than $10\,\text{g}$ of oxygen.

For centuries the evidence for individual atoms was clever inference from large-scale measurements. Scanning tunnelling microscopes can now resolve objects of atomic and molecular size.

But what if... we now have four quantities, p, V, T and n, for one gas? We must say what they describe together and how they are related: the **state variables** and the **equation of state**.



State variables and equation of state

Define: pressure p, volume V, temperature T and amount of substance n describe the conditions, or **state**, of the gas. These quantities are called **state variables**. The relationship among p, V, T and n is the **equation of state**.

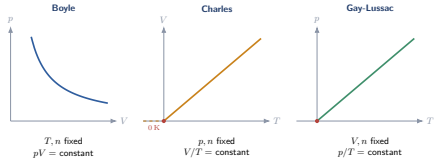
Explain. Ordinarily we cannot change one variable without changing another: fix three and the fourth follows.

Physical meaning

Large-scale measurements (p, V, T, n) describe the state of the gas without following any molecule. Lecture 9 gave the meaning of T; this lecture connects p and T to molecules.

But what if... we do not yet know the equation of state of a gas? Let experiments find it: change one variable, hold the others fixed, and watch the third.

Three experiments with a fixed amount of gas



Rule: three experiments, fixed amount of gas

Boyle (T, n fixed): pV is constant. Charles (p, n fixed): V/T is constant, with T in kelvin. Gay-Lussac (V, n fixed): p/T is constant.

Predict first

Charles: a line through the origin only if T is in kelvin. If V is $2.0\,\mathrm{L}$ at $300\,\mathrm{K}$ and p is fixed, what is V at $600\,\mathrm{K}$? And at $600^\circ\mathrm{C}$?

At $600\,\mathrm{K}$: $4.0\,\mathrm{L}$. At $600^\circ\mathrm{C} = 873\,\mathrm{K}$: $5.8\,\mathrm{L}$, *not* $4.0\,\mathrm{L}$. Doubling the Celsius number does not double the volume.

But what if... the three experiments are only three separate rules? First test your understanding of them, then combine them into one equation.

Decide first

1. At constant pressure, does doubling the Celsius temperature double the volume?
2. Does a gas have a fixed volume of its own?
3. Two containers of equal volume hold different gases at the same p and T. Do they hold the same number of molecules?
4. Is a pressure gauge reading of zero the same as zero pressure?

Answers

1. Doubling the Celsius temperature doubles the volume? **No.** Only the kelvin temperature is proportional to the volume.
2. A gas has a fixed volume of its own? **No.** It fills its container.
3. Equal volumes of different gases at the same p and T hold the same number of molecules? **Yes** (Avogadro's law: $V \propto n$ at fixed p and T).
4. A gauge reading of zero is zero pressure? **No.** A gauge reads $p - p_{\text{atm}}$, so zero gauge pressure is one atmosphere.

But what if... the three rules and $V \propto n$ can be combined? One equation should hold all four: the **ideal-gas equation**.

Rule: ideal-gas equation

$$pV = nRT = NkT$$

$R = 8.314\,\mathrm{J/(mol\,K)}$ is the gas constant; $k = R/N_A = 1.381 \times 10^{-23}\,\mathrm{J/K}$ is the gas constant per molecule (the Boltzmann constant). p is the **absolute** pressure and T the absolute (**kelvin**) temperature.

Physical meaning

Boyle, Charles and Gay-Lussac are this one equation with one variable held fixed: pV constant at fixed T; V/T constant at fixed p; p/T constant at fixed V.

But what if... we use the equation for any gas at any pressure? It is a *model*. We must say exactly what an ideal gas is, and when it can be trusted.

Definition: ideal gas

An **ideal gas** is one for which $pV = nRT$ holds precisely for all pressures and temperatures. In molecular terms, it is a gas whose molecules exert no attractive forces on each other and therefore have no potential energy.

Model validity

This is an idealized model: it works best at very low pressure and high temperature, when the molecules are far apart and in rapid motion. It is valid within a few percent at moderate pressures (a few atmospheres) and at temperatures well above those at which the gas liquefies. Real gases deviate at high pressure and near the liquid state (later in Contact A).

But what if... we put numbers into the equation in $^\circ\mathrm{C}$, in litres or with a gauge pressure? The answer is nonsense. We need a **unit habit** first: Toolkit 1.

- **TOOLKIT 1 – State variables and units**

1. **LIST** what is given: p, V, T, and either n, m_{total} or N. Which one is unknown?
2. **TEMPERATURE** in kelvin: $T = T_C + 273.15$ (never $^\circ\text{C}$ in a gas law).
3. **PRESSURE** in pascals, *absolute*: $1\,\text{atm} = 1.013 \times 10^5\,\text{Pa}$; a gauge reads $p_{\text{gauge}} = p - p_{\text{atm}}$.
4. **VOLUME** in m^3 ($1\,\text{L} = 10^{-3}\,\text{m}^3$); **AMOUNT** in moles, with M in kg/mol.

WATCH OUT: using $^\circ\text{C}$, using a gauge pressure, or M in g/mol with R in J/(mol K).
CHECK: units: $\text{Pa}\,\text{m}^3 = \text{J}$, so pV and nRT are both energies.

But what if... the units are right but we are unsure what each symbol in the equation stands for? Read the equations symbol by symbol: **equation anatomy**.

EQUATION – ideal-gas equation

$$pV = nRT = NkT$$

Meaning. relates the absolute pressure, volume, absolute temperature and amount of an ideal gas (its equation of state).

Variables. p absolute pressure; V volume; T kelvin; n moles; N molecules.

Units. $\mathrm{Pa\,m^3 = J}$ on both sides.

Interpretation. at fixed n, $T \propto pV$; it contains Boyle's, Charles's and Gay-Lussac's laws.

Example. the helium tank: $n = 1.83\,\mathrm{mol}$.

EQUATION – amount of substance

$$m_{\text{total}} = nM, \qquad N = nN_A, \qquad M = N_A m$$

Meaning. the mass and the number of molecules from the number of moles.

Variables. m_{total} total mass; M molar mass in kg/mol; m mass of one molecule; $N_A = 6.022 \times 10^{23}\,\mathrm{mol^{-1}}$.

Units. kg; mol.

Interpretation. equal moles, not equal masses, mean equal numbers of molecules.

Example. $m(\mathrm{N_2}) = 4.65 \times 10^{-26}\,\mathrm{kg}$.

But what if... we know the symbols but need a reliable routine for a real problem? Collect the steps in **Toolkit 2**.

- **TOOLKIT 2 – The ideal-gas equation**

1. IDENTIFY one state of the gas: three of p, V, T, n known.
2. CHOOSE the form: $pV = nRT$ (moles) or $pV = NkT$ (molecules); with $m_{\text{total}} = nM$ it gives the density $\rho = m_{\text{total}}/V = pM/(RT)$.
3. SOLVE for the unknown, then convert: $N = nN_A$, $m_{\text{total}} = nM$.
4. PREDICT first: at standard temperature and pressure (STP, $0^\circ\text{C} = 273.15\,\text{K}$ and $1\,\text{atm} = 1.013 \times 10^5\,\text{Pa}$) one mole occupies about $22.4\,\text{L}$.

WATCH OUT: use this form only for *one state*. For a *change* of state use the ratio method (Toolkit 3).
CHECK: is the number of moles of a sensible size for the container?

But what if... we try the routine on a real gas cylinder, predicting the size of the answer first? → **Example 1**.

Question

A tank of volume $15.0\,\mathrm{L}$ holds helium at an absolute pressure of $3.0 \times 10^5\,\mathrm{Pa}$ and a temperature of $22^\circ\mathrm{C}$. Find (a) the number of moles, (b) the number of atoms, (c) the mass of helium.

Predict first

A mole of gas at $1\,\mathrm{atm}$ fills about $22\,\mathrm{L}$. This tank is smaller but at about $3\,\mathrm{atm}$: about 0.2, 2 or 20 moles?

Set up: $V = 15.0 \times 10^{-3}\,\mathrm{m}^3$, $T = 295.15\,\mathrm{K}$.

Execute: (a) $n = \dfrac{pV}{RT} = \dfrac{(3.0 \times 10^5)(15.0 \times 10^{-3})}{(8.314)(295.15)} = 1.83\,\mathrm{mol}$. (b) $N = nN_A = 1.10 \times 10^{24}$. (c) $m_{\text{total}} = nM = (1.83)(4.00\,\mathrm{g/mol}) = 7.3\,\mathrm{g}$.

Evaluate: about 2 moles, as predicted; $7.3\,\mathrm{g}$ of helium is a plausible mass for a small tank. ✓

But what if... the gas is squeezed or heated, so the final state is not given by one set of numbers? Compare the two states directly: the **ratio method**.

Rule: fixed amount of gas

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \qquad (n \text{ constant}).$$

Same units on both sides; T in kelvin; p absolute.

Physical meaning

For a fixed amount of gas nR is constant, so pV/T has the same value in every state. To compare two states of the gas you need neither n nor the value of R.

But what if... we apply it without knowing what each symbol means and what is held fixed? Read it symbol by symbol: **equation anatomy**.

EQUATION – change of state, fixed amount

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$$

Meaning. for a fixed amount of gas, the combination pV/T stays constant.

Variables. subscripts 1 and 2 label the two states; T kelvin; p absolute.

Units. the units of p and V cancel in the ratio, but T must be in kelvin.

Interpretation. cancel whatever is constant: Boyle (T), Charles (p), Gay-Lussac (V).

Example. the pump: $5.78\,\mathrm{atm}$.

Predict first

The volume of a gas halves while its kelvin temperature doubles. Predict what happens to the pressure before using the equation.

$\frac{p_2}{p_1} = \frac{V_1}{V_2}\frac{T_2}{T_1} = 2 \times 2 = 4$: the pressure quadruples.

But what if... we want a step-by-step routine for such problems? Collect the steps in **Toolkit 3**.

TOOLKIT 3 – The ratio method for a change of state

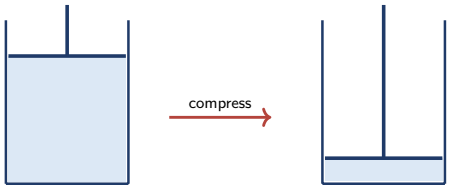
1. **STATES:** write the numbers for state 1 and state 2; mark the unknown.
2. **WHAT IS CONSTANT?** T (Boyle), p (Charles), or V (Gay-Lussac): cancel it.
3. **SOLVE** $p_2 = p_1 \dfrac{V_1}{V_2}\dfrac{T_2}{T_1}$ (or for the other unknown). Units of p and V cancel in the ratio.

WATCH OUT: n must not change (no leaks); T in kelvin. **CHECK:** does each ratio push the answer the way you predicted?

But what if... we test the routine on the bicycle pump we started with? → **Example 2**: the opening puzzle in numbers.

Example 2: compressing air in a pump

CORE



Air at $1.00\,\mathrm{atm}$ and $15^\circ\mathrm{C}$ fills a closed pump barrel of volume $0.60\,\mathrm{L}$. It is compressed quickly to $0.12\,\mathrm{L}$, and its temperature rises to $60^\circ\mathrm{C}$. Find the final pressure.

Predict first

The volume falls by a factor of 5. Is the pressure exactly $5\,\mathrm{atm}$, more, or less?

Question, restated

Air at $1.00\,\mathrm{atm}$ and $15^\circ\mathrm{C}$ fills a closed pump barrel of volume $0.60\,\mathrm{L}$. It is compressed quickly to $0.12\,\mathrm{L}$ and warms to $60^\circ\mathrm{C}$. Find the final pressure. Is it exactly $5\,\mathrm{atm}$, more, or less?

Set up: nothing leaks, so n is fixed. $T_1 = 288.15\,\mathrm{K}$, $T_2 = 333.15\,\mathrm{K}$.

Execute: $p_2 = p_1 \dfrac{V_1}{V_2}\dfrac{T_2}{T_1} = (1.00\,\mathrm{atm})\left(\dfrac{0.60}{0.12}\right)\left(\dfrac{333.15}{288.15}\right) = 5.78\,\mathrm{atm} = 5.9 \times 10^5\,\mathrm{Pa}.$

Evaluate: more than $5\,\mathrm{atm}$, as predicted: the compression raises the pressure by 5, and the warming adds a further 16%. So p does not rise by the same factor as V falls, because T rises too (the opening puzzle). Using $^\circ\mathrm{C}$ would give the absurd ratio $60/15 = 4$. ✓

But what if... we are given only the changes and must find the temperature, with answer choices that tempt a Celsius error? A concept check.

Choose, then explain

A fixed amount of ideal gas is at $27^\circ\mathrm{C}$. Its pressure is halved and its volume is tripled. What is the new temperature?

A. $177^\circ\mathrm{C}$ B. $27^\circ\mathrm{C}$ C. $40.5^\circ\mathrm{C}$ D. $-123^\circ\mathrm{C}$ E. $-177^\circ\mathrm{C}$

Predict first

Does the product pV go up or down? What does that say about T (before any arithmetic)?

pV is multiplied by $\frac{1}{2} \times 3 = 1.5$, so T rises by the same factor: $T_2 = 1.5 \times 300.15 = 450\,\mathrm{K} = \mathbf{177}^\circ\mathrm{C}$ (A). Answer C comes from multiplying the Celsius number: the common error.

But what if... we so far compared states with arithmetic, one pair at a time? One picture can show every state of a fixed amount of gas and tell us which is hotter: the **pV diagram**.

Sketch first

A fixed amount of ideal gas can be at many values of p and V. On a graph with V along the horizontal axis and p upward, sketch the curve joining all states at one temperature T_1. Then add the curve for a higher temperature T_2.

Predict first

Does the curve touch an axis? Does the higher-temperature curve lie closer to the origin or farther from it?

The curve $p = nRT/V$ is a hyperbola: it never touches an axis, and a higher temperature gives a curve farther from the origin. Each such curve is an **isotherm**.

But what if... we can draw the curves but cannot yet say which state is hotter, or what a curve means? The **pV diagram** turns the ideal-gas equation into a rule.

Rule: comparing states on a pV diagram

$$pV = nRT \quad \Rightarrow \quad \text{for the same } n: \; T \propto pV$$

The product pV (height times width of the rectangle under the point) tells the temperature. Equal pV: the same isotherm, the same temperature.

- A **pV-diagram** plots pressure p against volume V. Each curve, for one specific constant temperature, is an **isotherm** (a pV-isotherm): along it pV is constant (Boyle's law).
- A vertical path is constant V; a horizontal path is constant p.
- Only the ideal gas gives exact hyperbolas (real gases follow below).

But what if... a state with high p and small V can be as hot as one with low p and large V? Then looking at p alone, or V alone, misleads. We need a routine for comparing states: **Toolkit 4**.

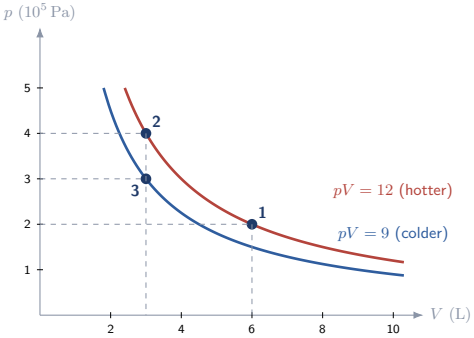
- **TOOLKIT 4 – Reading pV and phase diagrams: the pV diagram**

1. AXES: p against V. What is fixed (the amount of gas)?
2. pV DIAGRAM: sketch the isotherm through each point; the farther from the origin, the higher the T. Compare the products pV.
3. PATHS: horizontal = constant p; vertical = constant V; hyperbola = constant T.
WATCH OUT: comparing temperatures from p alone or V alone. **CHECK:** does the state on the higher isotherm have the larger product pV?

But what if... we use the routine on three states of one gas? → **Example 3**: which state is hottest?

Example 3: which state is hottest?

CORE



$0.50\,\text{mol}$ of an ideal gas can be in three states:

1: $p = 2.0 \times 10^5\,\text{Pa}$, $V = 6.0\,\text{L}$;
2: $p = 4.0 \times 10^5\,\text{Pa}$, $V = 3.0\,\text{L}$;
3: $p = 3.0 \times 10^5\,\text{Pa}$, $V = 3.0\,\text{L}$.

Which state is at the highest temperature? Find all three temperatures.

Predict first

Sketch the isotherms. Is one state clearly hottest, or are two tied?

Question, restated

$0.50\,\text{mol}$ of an ideal gas can be in three states: 1: $2.0\times10^5\,\text{Pa}$, $6.0\,\text{L}$; 2: $4.0\times10^5\,\text{Pa}$, $3.0\,\text{L}$; 3: $3.0\times10^5\,\text{Pa}$, $3.0\,\text{L}$. Which state is at the highest temperature? Find all three temperatures.

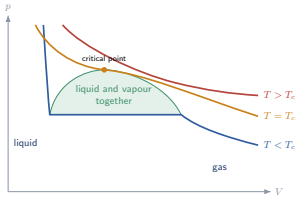
Set up: $T = pV/(nR)$; $nR = (0.50)(8.314) = 4.157\,\text{J/K}$.

Execute: $p_1V_1 = (2.0\times10^5)(6.0\times10^{-3}) = 1200\,\text{J}$,
$p_2V_2 = (4.0\times10^5)(3.0\times10^{-3}) = 1200\,\text{J}$, $\quad p_3V_3 = 900\,\text{J}$.
$T_1 = T_2 = 1200/4.157 = 289\,\text{K}\ (16^\circ\text{C})$; $\quad T_3 = 900/4.157 = 217\,\text{K}\ (-57^\circ\text{C})$.

Evaluate: states 1 and 2 are tied for the highest temperature and lie on the same isotherm; state 3 lies closer to the origin and is colder, as the sketch showed. State 2 has the largest p and state 1 the largest V, yet they are tied: compare the product pV, not p or V alone. ✓

But what if... we keep compressing the gas, and the ideal-gas hyperbola says the pressure rises without limit? A real gas condenses first. How do real isotherms differ?

Real gases: the isotherms bend



Explain. At high density or low T the ideal-gas equation fails. Along the flat section the pressure is constant, the vapour condenses and the volume shrinks.

Real isotherms and the critical temperature T_c

Above T_c no liquid-vapour transition occurs and the isotherms resemble those of the ideal gas. Below T_c, compression at constant T reaches a flat section where liquid and vapour coexist.

A better model. The van der Waals equation corrects for the volume of the molecules and the attraction between them: $\left(p + \frac{an^2}{V^2}\right)(V - nb) = nRT$, with constants a and b for each gas.

But what if... the vapour turns into liquid and the two coexist? We need exact words for these forms of matter and the changes between them: the **vocabulary of phases**.

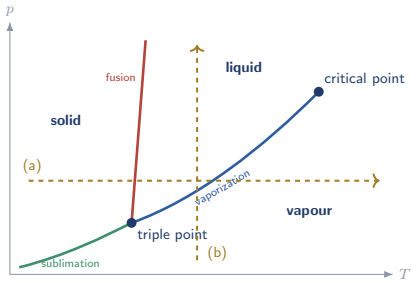
Defining the vocabulary of phases

CORE

Quantity	What it is	What it measures	SI unit	Representation
Phase	a specific state of matter: solid, liquid or gas (vapour)	how the molecules are arranged	—	a region of the phase diagram
Phase change	a transition from one phase to another, at a definite temperature for a given pressure, with heat flowing in or out	when matter switches phase	—	a flat section of the heating curve
Phase equilibrium	two phases coexisting; at a given pressure this happens at one specific temperature	when two phases can coexist	—	a point on a curve of the phase diagram

Interpret: a phase answers *which form is the substance in?*; a phase change answers *when does it switch?* Sublimation is the change directly from solid to gas.

But what if... we want to know which phase exists at a given pressure and temperature? A map of p against T: the **phase diagram**.



Rule: reading a phase diagram

Each point (T, p) shows one phase, except on the **fusion**, **vaporization** and **sublimation** curves, where two phases coexist in phase equilibrium. All three coexist only at the **triple point**. The vaporization curve ends at the **critical point** (T_c, p_c), where the distinction between liquid and vapour disappears.

Path (a): heating at constant p. Path (b): compression at constant T. Boiling and melting temperatures depend on pressure.

But what if... we meet such a map and are unsure how to read it? Complete **Toolkit 4**.

• TOOLKIT 4 – Reading pV and phase diagrams: the phase diagram

4. PHASE DIAGRAM: p against T. Read the region (solid, liquid, vapour); a path that crosses a curve is a change of phase; on a curve two phases coexist in phase equilibrium.
5. SPECIAL POINTS: triple point (all three phases coexist); critical point (the end of the vaporization curve: no distinction between liquid and vapour).

WATCH OUT: a phase diagram is p against T; the pV diagram of a real gas has a liquid-vapour region. **CHECK:** does the path cross the curves you expect?

But what if... we try the map on a real substance? → **Example 4**: dry ice at normal pressure.

Question

The triple point of carbon dioxide is at $216.55\,\mathrm{K}$ and $5.17 \times 10^5\,\mathrm{Pa}$. A block of solid $\mathrm{CO_2}$ (dry ice) is left in a room at $1\,\mathrm{atm}$. Does it melt to a liquid? What would you have to do to obtain liquid $\mathrm{CO_2}$?

Predict first

Sketch a horizontal line at $1\,\mathrm{atm}$ on the phase diagram. Is $1\,\mathrm{atm}$ above or below the triple-point pressure?

$1\,\mathrm{atm} = 1.013 \times 10^5\,\mathrm{Pa}$ is about 0.2 of the triple-point pressure ($5.10\,\mathrm{atm}$), so the line passes below the triple point: it meets only the sublimation curve. The solid **sublimes** directly to vapour (at about $-78^\circ\mathrm{C}$). To obtain the liquid, the pressure must exceed $5.17 \times 10^5\,\mathrm{Pa}$.
Evaluate: this is why dry ice leaves no puddle. ✓

But what if... the dry ice takes in heat from the room, yet it does not warm up and does not melt either? That is the first question of Contact B. First collect the slips of Contact A.

Common mistakes and the correct habit

Mistake	Instead
Using $^\circ$C or gauge pressure in a gas law.	Kelvin and absolute pressure.
Assuming equal masses mean equal numbers of molecules.	Equal moles do.
Using the one-state equation for a change of state.	Use the ratio method with n fixed.
Doubling the Celsius number to double the volume.	Only T in kelvin is proportional to V.
Using M in g/mol with R in J/(mol K).	Use kg/mol.
Comparing temperatures from p alone or V alone.	Compare the product pV, or the isotherms.
Confusing the phase diagram with the pV diagram.	A phase diagram is p against T.

Bridge: what we can now do, and the questions for Contact B

CORE

Contact A: you should be able to

State variables p, V, T, n ($N = nN_A$) $\Rightarrow$ **equation of state** $pV = nRT = NkT \Rightarrow$ **change of state** $\dfrac{p_1V_1}{T_1} = \dfrac{p_2V_2}{T_2}$.

- Convert to kelvin, absolute pressure and m^3; use $pV = nRT$ and the ratio method; say when the ideal-gas model is valid.
- Compare states on a pV diagram ($T \propto pV$).
- Read a phase diagram, including the triple and critical points.

Predict first

Before Wednesday, predict: a pan of water is boiling on the stove. If you turn the heat up, does the water get hotter? And what is it, inside a sealed container, that presses on the walls of a gas?

Connect: Lectures 11 and 12

A path on the pV diagram will give the work done by the gas (Lecture 11); closed paths are engine cycles (Lecture 12).

But what if... ice at $0^\circ\mathrm{C}$ takes in heat without warming, and a gas presses on its walls? $\rightarrow$ **Contact B**.

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Contact B: Analyse, then Explain

What does heat do at a change of phase, and what do molecules do to give pressure and temperature?

Predict first

1. A fixed amount of gas at $300\,\mathrm{K}$ and $1.0\,\mathrm{atm}$ is sealed in a rigid container and heated to $600\,\mathrm{K}$. What is the new pressure? V and n are fixed, so $p_2 = p_1 T_2/T_1 = 2.0\,\mathrm{atm}$ (kelvin, ratio method).
2. Two states of one gas have the same product pV on a pV diagram. Are they at the same temperature? Yes: $T = pV/(nR)$, so they lie on the same isotherm.
3. How much heat raises $0.50\,\mathrm{kg}$ of water by $20\,\mathrm{K}$ ($c = 4186\,\mathrm{J/(kg\,K)}$)? $Q = mc\Delta T = (0.50)(4186)(20) = 4.2 \times 10^4\,\mathrm{J}$.

But what if... ice at $0^\circ\mathrm{C}$ takes in heat but does not warm? Today: heat at a change of phase, then calorimetry, then what the molecules are doing.

Decide first

1. You heat ice at $0^\circ\mathrm{C}$. What happens to its temperature while it melts?
2. Does it take more energy to melt $1\,\mathrm{kg}$ of ice or to boil away $1\,\mathrm{kg}$ of water?
3. A pan of boiling water is on the stove at full power. Does the water get hotter if the power is increased?

Answers

1. Temperature of ice at $0^\circ\mathrm{C}$ while it melts? **It stays at $0^\circ\mathrm{C}$.** The heat changes the phase, not the temperature: the energy goes into the arrangement of the molecules, not into their kinetic energy.
2. More energy to melt $1\,\mathrm{kg}$ of ice or to boil $1\,\mathrm{kg}$ of water? **Boiling,** about seven times as much: vaporization separates the molecules completely.
3. Hotter if the power is increased? **No.** It boils faster; the temperature stays at the boiling point.

But what if... we need a number for this energy? Heat per kilogram of material: the **heat of fusion** and the **heat of vaporization**.

Defining the vocabulary of latent heat

CORE

Quantity	What it is	What it measures	SI unit	Representation
Heat of fusion L_f	the heat per unit mass to change solid to liquid at the melting temperature	energy cost of melting per kg	J/kg	flat section at the melting point
Heat of vaporization L_v	the heat per unit mass to change liquid to vapour at the boiling temperature	energy cost of boiling per kg	J/kg	flat section at the boiling point
Latent heat L	the heat per unit mass for a phase change, so $Q = \pm mL$ (L_s: sublimation)	energy cost of a phase change	J/kg	any flat section of the heating curve

Interpret: latent heat answers *how much energy does the change of phase cost per kilogram, with no change of temperature?*

But what if... the words do not tell us how much heat a given mass needs, or its sign? → the **rule** $Q = \pm mL$.

Rule: latent heat

$$Q = \pm mL$$

m: mass of material that changes phase. $+$: heat enters (melting, boiling); $-$: heat leaves (freezing, condensing). L in J/kg.

Physical meaning

During the change the temperature stays *constant*: the energy goes into changing the arrangement of the molecules, not into raising their average kinetic energy.

Substance	L_f (J/kg)	L_v (J/kg)
water	3.34×10^5	2.256×10^6
ethanol	1.04×10^5	8.54×10^5
nitrogen	2.55×10^4	2.01×10^5

Normal pressure. $L_v > L_f$: vaporization separates molecules completely.

But what if... a process has a temperature change and a phase change? Which equation applies where? Read both side by side: **equation anatomy**.

EQUATION – within one phase

$$Q = mc\,\Delta T$$

Meaning. heat to change the temperature of a mass m without a change of phase.

Variables. c specific heat of *that* phase; $\Delta T = T_{\text{final}} - T_{\text{initial}}$.

Units. J.

Interpretation. a slanted section of the heating curve.

Example. warming $0.200\,\text{kg}$ of ice by $15\,\text{K}$: $6.3 \times 10^3\,\text{J}$.

EQUATION – at a change of phase

$$Q = \pm mL$$

Meaning. heat transferred in a change of phase of a mass m, at constant temperature.

Variables. L heat of fusion L_f or of vaporization L_v (latent heat); $+$ if heat enters, $-$ if heat leaves.

Units. $\text{kg} \cdot \text{J/kg} = \text{J}$.

Interpretation. a flat section of the heating curve.

Example. melting the ice: $6.68 \times 10^4\,\text{J}$.

But what if... a real process runs through several stages in a row (warm, melt, warm)? We need a routine that adds the stages: **Toolkit 5**.

- **TOOLKIT 5 – Heating curves and latent heat**

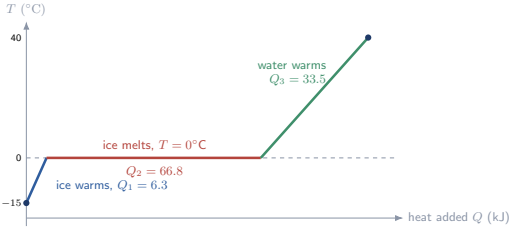
1. SKETCH the heating curve (T against Q, or against time at constant power): slanted lines in one phase, flat lines at a change of phase.
2. LIST THE STAGES from the initial to the final state (warm the solid, melt, warm the liquid, boil, warm the vapour).
3. SLANTED STAGE: $Q = mc\Delta T$ with c of *that* phase (Lecture 9).
4. FLAT STAGE: $Q = \pm mL$ at the transition temperature.
5. ADD the stages (all positive when heating); $Q = Pt$ turns energy into time.

WATCH OUT: using one c for all phases; forgetting a stage; using $mc\Delta T$ at a change of phase.
CHECK: does the largest bar on the sketch correspond to the largest Q?

But what if... we apply it from cold ice to warm water? → **Example 5**.

Example 5: from ice to warm water

CORE



How much heat is needed to take $0.200\,\mathrm{kg}$ of ice at $-15^\circ\mathrm{C}$ to water at $40^\circ\mathrm{C}$? ($c_{\text{ice}} = 2100$, $c_{\text{water}} = 4186\,\mathrm{J/(kg\,K)}$, $L_f = 3.34 \times 10^5\,\mathrm{J/kg}$.)

Predict first

Which stage takes the most heat: warming the ice, melting it, or warming the water? Why?

Question, restated

How much heat takes $0.200\,\mathrm{kg}$ of ice at $-15^\circ\mathrm{C}$ to water at $40^\circ\mathrm{C}$? ($c_{\mathrm{ice}} = 2100$, $c_{\mathrm{water}} = 4186\,\mathrm{J/(kg\,K)}$, $L_f = 3.34 \times 10^5\,\mathrm{J/kg}$.) Which stage takes the most heat?

Execute: $Q_1 = mc_{\mathrm{ice}}\Delta T = (0.200)(2100)(15) = 6.3 \times 10^3\,\mathrm{J}$.
$Q_2 = mL_f = (0.200)(3.34 \times 10^5) = 6.68 \times 10^4\,\mathrm{J}$.
$Q_3 = mc_{\mathrm{water}}\Delta T = (0.200)(4186)(40) = 3.35 \times 10^4\,\mathrm{J}$.
$Q = Q_1 + Q_2 + Q_3 = 1.07 \times 10^5\,\mathrm{J}$.

Evaluate: melting takes the most (63%), as the flat line of the sketch is the longest, even though it involves no temperature change. All terms are positive, as heat enters. ✓

But what if... the heat does not come from a heater but from another body in the same container, such as warm water around ice? We need a rule for heat exchanged between bodies: **calorimetry**.

Rule: calorimetry

When heat flows between bodies that are isolated from their surroundings, the heat lost by one body equals the heat gained by the other. Take $Q > 0$ for heat entering a body and $Q < 0$ for heat leaving it:

$$\sum Q = 0, \qquad Q = mc\,\Delta T \text{ (within a phase)}, \qquad Q = \pm mL \text{ (at a change of phase)},$$

with $\Delta T = T_{\text{final}} - T_{\text{initial}}$.

Physical meaning

Heat is energy in transit, so this rule is conservation of energy: calorimetry ("measuring heat") rests on one conserved quantity.

But what if... ice is one of the bodies, and we do not know whether all of it melts, so we do not know which terms belong in the sum? Decide the final state first: **Toolkit 6**.

- **TOOLKIT 6 – Calorimetry with a change of phase**

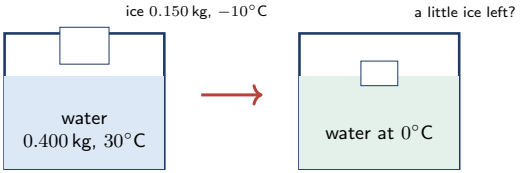
1. SYSTEM: insulated; list every body, its phase and initial temperature.
2. DECIDE THE FINAL STATE FIRST: compute the heat the warmer body can give up in cooling to the transition temperature, and the heat the colder body needs to reach it and to complete its change of phase. Compare.
3. WRITE $\sum Q = 0$ with the terms that belong to *that* final state ($mc\Delta T$ and $\pm mL$).
4. SOLVE for the unknown (T_f, or the mass that changes phase).
5. EVALUATE: the result must be consistent with the assumed state.

WATCH OUT: assuming the final state; sign of the mL terms; forgetting the warming of the ice. **CHECK:** is $0 \leq m_{\text{melted}} \leq m_{\text{ice}}$?

But what if... we test the routine in a close case, where a wrong guess shows? → **Example 6**: ice added to water.

Example 6: ice added to water

CORE



$0.150\,\mathrm{kg}$ of ice at $-10^\circ\mathrm{C}$ is dropped into $0.400\,\mathrm{kg}$ of water at $30^\circ\mathrm{C}$ in an insulated container. Find the final state.

Predict first

Sketch the heat flow. Will all the ice melt? Will the final temperature be above $0^\circ\mathrm{C}$, exactly $0^\circ\mathrm{C}$, or below?

Question, restated

$0.150\,\mathrm{kg}$ of ice at $-10^\circ\mathrm{C}$ is dropped into $0.400\,\mathrm{kg}$ of water at $30^\circ\mathrm{C}$ in an insulated container. Find the final state. Will all the ice melt? Will the final temperature be above, at or below $0^\circ\mathrm{C}$?

Decide the final state: heat available from the water cooling to $0^\circ\mathrm{C}$: $0.400(4186)(30) = 5.02 \times 10^4\,\mathrm{J}$.
Heat needed to warm the ice to $0^\circ\mathrm{C}$: $0.150(2100)(10) = 3.2 \times 10^3\,\mathrm{J}$; to melt all of it: $0.150(3.34 \times 10^5) = 5.01 \times 10^4\,\mathrm{J}$; total $5.33 \times 10^4\,\mathrm{J}$.

Needed $>$ available: not all the ice melts, and the final temperature is $0^\circ\mathrm{C}$.

But what if... we now need *how much* ice melts? With the final state fixed, the sum $\sum Q = 0$ has one unknown, the melted mass.

Question, restated

$0.150\,\mathrm{kg}$ of ice at $-10^\circ\mathrm{C}$ and $0.400\,\mathrm{kg}$ of water at $30^\circ\mathrm{C}$ in an insulated container. We found that the final temperature is $0^\circ\mathrm{C}$ with some ice left. How much ice melts, and what is the final state?

Execute: the water gives up $50\,232\,\mathrm{J}$; $3150\,\mathrm{J}$ warm the ice and the rest melts it:
$m_{\text{melted}} = \dfrac{50\,232 - 3150}{3.34 \times 10^5} = 0.141\,\mathrm{kg}$; ice left: $0.150 - 0.141 = 0.009\,\mathrm{kg}$.

Evaluate: final: $T = 0^\circ\mathrm{C}$ with $0.541\,\mathrm{kg}$ of water and $9\,\mathrm{g}$ of ice. The comparison was close (53.3 against $50.2\,\mathrm{kJ}$), so the check mattered: guessing "all melts" would give a result below $0^\circ\mathrm{C}$ for the water. ✓

But what if... the heat and phase change are settled, and the molecules are still out of sight? First a short checkpoint. (Steam into water and a path on the pV diagram are tutorial problems B14 and B15.)

Heat and phase change: what we can now do

latent heat: $Q = \pm mL$, heating curve
$\Downarrow$ *heat shared between bodies*
calorimetry: decide the final state, then $\sum Q = 0$

You should be able to

- calculate Q across a change of phase;
- solve a calorimetry problem with a phase change.

Connect: Lectures 11 and 12

Latent heat: during a change of phase the thermal energy changes at constant T (first law); melting increases the disorder of the molecules (second law).

A new question: what are the molecules doing?

So far we described matter with large-scale quantities (p, V, T, Q) and diagrams, without following a single molecule. We now ask what lies behind them.

Question	Tool in the second half of Contact B
Where does the pressure of a gas come from?	the kinetic model and collisions with the wall
What does temperature mean for a molecule?	the average kinetic energy $\frac{3}{2}kT$
How fast do molecules move, and do they all move equally fast?	the rms speed and the speed distribution

But what if... we begin with the first question, before any theory? Ask what a gas sealed in a container does to its walls.

Poll

A gas in a sealed container presses on the walls. Nothing is pushing the gas against the walls. **What causes the pressure?** If every molecule moved twice as fast, by what factor would the pressure change?

Predict first

Sketch one molecule bouncing off a wall. Faster molecules hit harder, and they also hit more often. Does the pressure double, quadruple, or stay the same?

Pressure is the magnitude of the force per unit area; the force comes from the very many collisions of molecules with the wall. Each collision is $2\times$ stronger and the collisions are $2\times$ as frequent: the pressure becomes $\mathbf{4}\times$. We now make this precise.

But what if... the factor 4 came from two loose arguments in words? To make it exact we must say precisely what the molecules are assumed to do: the **kinetic model**.

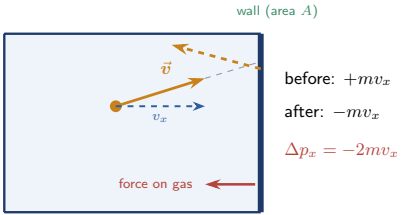
Assumption	Where it is used
N identical molecules, each of mass m, in a container of volume V (N very large)	averages are meaningful
The molecules move randomly, with no preferred direction	the same average for each direction: $(v_x^2)_{\text{av}} = (v_y^2)_{\text{av}} = (v_z^2)_{\text{av}}$
Point particles, small against the container and their separation	the whole container volume is free
No attractive forces act between the molecules	no potential energy; straight-line paths
Perfectly elastic collisions with rigid, infinitely massive walls	the speed is unchanged, $v_x \to -v_x$

Physical meaning

Each assumption is a simplification. The model works best for low density, which is exactly where the ideal-gas equation works.

But what if... we want the force that this gas puts on a wall? Follow **one molecule** bouncing between two walls.

One molecule, one wall



Change in momentum at the wall

A molecule in a cubical box of side L hits the right wall elastically. Its momentum changes by $\Delta P_x = -2mv_x$ (capital P for momentum, small p for pressure).

- It returns to the same wall after a time $2L/v_x$.
- Average force on the wall:
$$F = \frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}.$$

The force on the wall is the rate of change of momentum (Lecture 7), with the sign reversed by Newton's third law.

But what if... one molecule gives only a tiny, intermittent force? Add up all N molecules and divide by the wall area to get the **pressure**.

1. **Sum over N molecules:** $F = \dfrac{Nm\,(v_x^2)_{\text{av}}}{L}$.
2. **Three directions are equivalent:** $(v^2)_{\text{av}} = 3(v_x^2)_{\text{av}}$, so $(v_x^2)_{\text{av}} = \frac{1}{3}(v^2)_{\text{av}}$.
3. **Pressure is force per area** $(A = L^2)$: $p = \dfrac{F}{A} = \dfrac{Nm(v^2)_{\text{av}}}{3L^3} = \dfrac{1}{3}\dfrac{N}{V}m(v^2)_{\text{av}}$.

Rule: pressure of an ideal gas

$$pV = \tfrac{1}{3}Nm\,(v^2)_{\text{av}} = \tfrac{2}{3}N\left[\tfrac{1}{2}m(v^2)_{\text{av}}\right]$$

The pressure grows with the number density and with the average translational kinetic energy per molecule.

But what if... we want to read the result symbol by symbol and check it against the opening question? **Equation anatomy**.

EQUATION – kinetic pressure

$$pV = \tfrac{1}{3} N m \, (v^2)_{\text{av}}$$

Meaning. pressure is the magnitude of the force per unit area; the force comes from the collisions of molecules with the wall.

Variables. N molecules; m mass of one molecule; $(v^2)_{\text{av}}$ average of the squared speeds.

Units. $\text{Pa}\,\text{m}^3 = \text{J}$.

Interpretation. more molecules or faster ones mean more pressure; doubling v quadruples p.

Example. the opening question: every molecule twice as fast, same N and V: $(v^2)_{\text{av}} \to 4(v^2)_{\text{av}}$, so $p \to 4p$.

Predict first

Predict what happens to the pressure if the number of molecules doubles at fixed V and v.

$pV \propto N$ at fixed V, m and $(v^2)_{\text{av}}$: the pressure doubles.

But what if... the combination $\frac{1}{2} m (v^2)_{\text{av}}$ keeps appearing, and so does a typical speed? They need names: the **vocabulary of kinetic theory**.

Defining the vocabulary of kinetic theory

Quantity	What it is	What it measures	SI unit	Representation
Molecular kinetic energy K	the average translational kinetic energy of one molecule, $\frac{1}{2}m(v^2)_{\text{av}} = K_{\text{tr}}/N$ (K_{tr}: total for all N)	energy of an average molecule	J	depends only on T
rms speed v_{rms}	the root of the mean of the squares, $\sqrt{(v^2)_{\text{av}}}$	a typical molecular speed	m/s	a point on the speed distribution
Boltzmann constant k	R/N_A, the gas constant on a per-molecule basis	the energy scale per molecule per kelvin	J/K	$1.381 \times 10^{-23}\,\text{J/K}$

Interpret: K answers *how much translational energy does an average molecule carry?* and v_{rms} answers *how fast is a typical molecule?*

But what if... the kinetic result and the ideal-gas equation both give pV? Compare them: what does **temperature** mean for a molecule?

Rule: kinetic energy and temperature

Compare $pV = \frac{2}{3}N\left[\frac{1}{2}m(v^2)_{\text{av}}\right]$ with $pV = NkT$:

$$K = \tfrac{1}{2}m(v^2)_{\text{av}} = \tfrac{3}{2}kT \qquad K_{\text{tr}} = NK = \tfrac{3}{2}nRT$$

For one mole: $N_A K = \frac{1}{2}M(v^2)_{\text{av}} = \frac{3}{2}RT$, with $M = N_A m$.

Physical meaning

The average translational kinetic energy of a molecule is directly proportional to the absolute temperature. It depends only on T, not on p, V or the kind of molecule.

But what if... the average energy fixes a typical speed, but which average of the speeds? Because K contains v^2, the natural one is the **rms speed**.

Rule: rms speed

$$v_{\text{rms}} = \sqrt{(v^2)_{\text{av}}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

To find it, square each molecular speed, add, divide by the number of molecules and take the square root.

The mean of the *squares* is what appears in the kinetic energy, so v_{rms} follows directly; the average speed v_{av} is slightly smaller. At a given T, molecules of different mass have the same average kinetic energy but different rms speeds.

But what if… the rms speed and the ordinary average are different numbers? Compare them by hand for four molecules.

Check by hand: rms and average speed

Check by hand

Four molecules have speeds 200, 400, 600 and $800\,\mathrm{m/s}$. Find v_{av} and v_{rms}.

$v_{\mathrm{av}} = 500\,\mathrm{m/s}$; $\quad (v^2)_{\mathrm{av}} = \dfrac{200^2 + 400^2 + 600^2 + 800^2}{4} = 3.0 \times 10^5\,\mathrm{m^2/s^2}$, so $v_{\mathrm{rms}} = 548\,\mathrm{m/s}$.

$v_{\mathrm{rms}} > v_{\mathrm{av}}$: squaring weights the fast molecules more.

But what if… we now hold two new equations, $K = \frac{3}{2}kT$ and v_{rms}, and want every symbol and unit clear? **Equation anatomy**.

Equation anatomy: kinetic theory

EQUATION – temperature and energy

$$K = \tfrac{1}{2}m(v^2)_{\text{av}} = \tfrac{3}{2}kT, \qquad K_{\text{tr}} = \tfrac{3}{2}nRT$$

Meaning. the average translational kinetic energy of a molecule is proportional to the absolute temperature.

Variables. m mass of a molecule; k Boltzmann constant; T kelvin; $K_{\text{tr}} = NK$ total for all molecules.

Units. J.

Interpretation. K depends only on T; the total depends also on the amount.

Example. 6.07×10^{-21} J at 20°C, for every gas.

EQUATION – rms speed

$$v_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

Meaning. the root of the average of the squared speeds: a typical molecular speed.

Variables. M in kg/mol; m in kg.

Units. m/s.

Interpretation. $\propto \sqrt{T}$ and $\propto 1/\sqrt{M}$: lighter molecules are faster.

Example. N_2: 511 m/s; He: 1.35×10^3 m/s.

But what if... we want a routine for any kinetic-theory problem? Collect the steps in **Toolkit 7**.

- **TOOLKIT 7 – Kinetic theory**

1. **SKETCH** the molecules and the wall; state the assumptions that apply.
2. **PER MOLECULE:** $K = \frac{3}{2}kT$ (energy in J); use $m = M/N_A$ for the molecular mass.
3. **SPEED:** $v_{\text{rms}} = \sqrt{3kT/m} = \sqrt{3RT/M}$ with M in **kg/mol**.
4. **TOTAL:** $K_{\text{tr}} = NK = \frac{3}{2}nRT$ depends on the *amount*.
5. **SCALING:** $v_{\text{rms}} \propto \sqrt{T}$ and $\propto 1/\sqrt{M}$; $K \propto T$ only. For ratios use kelvin.

WATCH OUT: M in g/mol; °C in T; confusing energy per molecule with total energy.
CHECK: is v_{rms} of order a few hundred m/s at room temperature?

But what if... we apply it to two real gases at one temperature? → **Example 7**.

Question

Find the average translational kinetic energy per molecule and the rms speed of (a) oxygen (O_2, $M = 32.0\,\mathrm{g/mol}$) and (b) helium ($M = 4.00\,\mathrm{g/mol}$), both at $27^\circ\mathrm{C}$. Also find the total random translational kinetic energy of one mole of either gas.

Predict first

Which has the greater kinetic energy per molecule? Which is faster, and by what factor ($\sqrt{32/4}$)?

$T = 27 + 273 = 300\,\mathrm{K}$. $\quad K = \frac{3}{2}kT = \frac{3}{2}(1.381 \times 10^{-23})(300) = 6.21 \times 10^{-21}\,\mathrm{J}$ (both gases).
One mole: $K_{\mathrm{tr}} = \frac{3}{2}nRT = \frac{3}{2}(1)(8.314)(300) = 3740\,\mathrm{J}$.

$$v_{\mathrm{rms}}(O_2) = \sqrt{\frac{3(8.314)(300)}{0.0320}} = 484\,\mathrm{m/s}; \quad v_{\mathrm{rms}}(\mathrm{He}) = \sqrt{\frac{3(8.314)(300)}{0.00400}} = 1.37 \times 10^3\,\mathrm{m/s}.$$

Evaluate: equal K, helium faster by $1368/484 = 2.83 = \sqrt{8}$, as predicted; speeds of order hundreds of m/s, comparable with the speed of sound. ✓

But what if... "equal K but different speeds" surprised us? Test it on another pair of gases.

Choose, then explain

Argon ($M = 39.9\,\mathrm{g/mol}$) and neon ($M = 20.2\,\mathrm{g/mol}$) are both at $300\,\mathrm{K}$. How do the rms speed and the average translational kinetic energy per molecule compare?

A. Argon has the larger K and the larger v_{rms}.
B. Neon has the larger v_{rms}; K is the same.
C. Neon has the larger v_{rms}; argon has the larger K.
D. K and v_{rms} are the same for both.
E. Neon has the larger K; argon has the larger v_{rms}.

Predict first

Which quantity depends on the mass of the molecule, and which does not?

B. $K = \frac{3}{2}kT$ is the same; $v_{\mathrm{rms}} \propto 1/\sqrt{m}$ is larger for the lighter neon, by $\sqrt{39.9/20.2} = 1.41$.

But what if... equal energy per molecule means equal total energy? Not for equal *masses*: count the molecules.

Choose, then explain

$10\,\mathrm{g}$ of hydrogen (H_2, $M = 2.02\,\mathrm{g/mol}$) and $10\,\mathrm{g}$ of oxygen (O_2, $M = 32.0\,\mathrm{g/mol}$) are at $300\,\mathrm{K}$. Which sample has the greater *total* translational kinetic energy, and by roughly what factor?

Predict first

Same T, so same energy per molecule. Which sample contains more molecules?

$K_{\mathrm{tr}} = \frac{3}{2}nRT = \frac{3}{2}\frac{m_{\mathrm{total}}}{M}RT.$

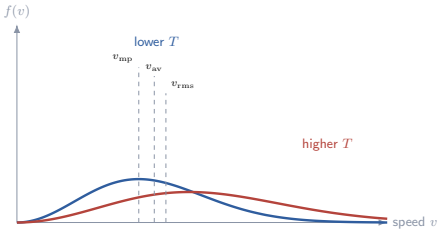
H_2: $n = 4.95\,\mathrm{mol}$, $K_{\mathrm{tr}} = 1.85 \times 10^4\,\mathrm{J}$. O_2: $n = 0.3125\,\mathrm{mol}$, $K_{\mathrm{tr}} = 1.17 \times 10^3\,\mathrm{J}$.

Evaluate: hydrogen has 15.8 times more, equal to M_{O_2}/M_{H_2}: same energy per molecule, but 15.8 times as many molecules. ✓

But what if... we have spoken of *the* speed and *the* energy of a molecule, as if all were alike? Do all the molecules really move equally fast?

Not all molecules move at the same speed

CORE



Speed distribution

The speeds form a **distribution**: a most probable speed and a long high-speed tail. Always $v_{\text{mp}} < v_{\text{av}} < v_{\text{rms}}$.

Raising T moves the peak to higher speed and widens the curve; the total area (all the molecules) is unchanged.

v_{rms} is only a representative speed: some molecules are much faster and some much slower. Formulas for the distribution are in the supplement.

But what if... a problem needs both the gas law and the kinetic model, linked by the temperature? → the **integrated problem**.



Application problem

A rigid sealed flask of volume $10.0\,\text{L}$ contains helium at $1.50\,\text{atm}$ and $20°\text{C}$. It is heated until the rms speed of the atoms is 25% larger.

(a) How many moles and atoms are there?
(b) Find the initial rms speed.
(c) Find the final temperature and pressure.

Predict first

Sketch the flask before and after. Will T rise by 25%, by more, or by less? Will the pressure rise by the same factor as T?

Question, restated

A rigid sealed flask of volume $10.0\,\mathrm{L}$ contains helium at $1.50\,\mathrm{atm}$ and $20^\circ\mathrm{C}$. It is heated until the rms speed of the atoms is 25% larger. (a) How many moles and atoms are there? (b) Find the initial rms speed. (c) Find the final temperature and pressure.

Identify, set up: a fixed n in a rigid flask: ideal gas (Toolkit 2) and the kinetic model (Toolkit 7), linked by T. $T_1 = 293.15\,\mathrm{K}$, $p_1 = 1.50 \times 1.013 \times 10^5 = 1.52 \times 10^5\,\mathrm{Pa}$, $V = 10.0 \times 10^{-3}\,\mathrm{m}^3$.

(a) $$n = \frac{p_1 V}{R T_1} = \frac{(1.52 \times 10^5)(10.0 \times 10^{-3})}{(8.314)(293.15)} = 0.623\,\mathrm{mol}; \qquad N = nN_A = 3.75 \times 10^{23}.$$

(b) $$v_{\mathrm{rms},1} = \sqrt{3RT_1/M} = \sqrt{\frac{3(8.314)(293.15)}{4.00 \times 10^{-3}}} = 1.35 \times 10^3\,\mathrm{m/s}.$$

But what if... the atoms must move 25% faster, and we do not yet know what that does to T and p? Part (c).

Question, restated

Part (c): the rigid sealed flask holds $0.623\,\text{mol}$ of helium at $293.15\,\text{K}$ and $1.52 \times 10^5\,\text{Pa}$ ($V = 10.0\,\text{L}$). It is heated until the rms speed of the atoms is 25% larger. Find the final temperature and pressure.

(c) $v_{\text{rms}} \propto \sqrt{T} \Rightarrow T_2 = (1.25)^2 T_1 = 1.5625\,T_1 = 458\,\text{K} = 185°\text{C}$. V is constant, so $p_2 = 1.5625\,p_1 = 2.34\,\text{atm} = 2.37 \times 10^5\,\text{Pa}$.

Evaluate: T rises by 56% (more than 25%), because v_{rms} depends on $\sqrt{T}$; p rises with T by the same factor, as predicted. Check with $pV = nRT$:
$p_2 = (0.623)(8.314)(458)/(10.0 \times 10^{-3}) = 2.37 \times 10^5\,\text{Pa}$. ✓

But what if... you leave with the model and its results? First collect the slips to avoid: the common mistakes of Contact B.

Common mistakes and the correct habit

Mistake	Instead
Using $Q = mc\Delta T$ across a change of phase.	Use $\pm mL$ at the change, $mc\Delta T$ within a phase.
Assuming the final state in calorimetry.	Decide the final state first by comparing the heats.
Using one c for all phases.	Use c of the phase you are in.
Confusing energy per molecule with total energy.	$K = \frac{3}{2}kT$ per molecule; $K_{\text{tr}} = \frac{3}{2}nRT$ in total.
Saying all molecules move at v_{rms}.	It is only a representative speed.
Using M in g/mol in v_{rms}.	Use kg/mol with R.
Saying temperature measures the total energy, or that molecules stop at 0°C.	The average kinetic energy per molecule is proportional to T; molecules stop only as $T \to 0\,\text{K}$.
Using the ideal-gas model at high density.	Check the low-density assumption.

But what if... these results were for a gas in a closed, rigid container? Where does the molecular picture lead next?

Lecture 11: first law

$K = \frac{3}{2}kT$ gives the thermal energy of a monatomic gas, $U = \frac{3}{2}nRT$; the area under a path on the pV diagram is the work done by the gas.

Lecture 12: engines

Engine cycles are closed paths on the pV diagram; the molecular picture explains why heat flows from hot to cold.

Applications

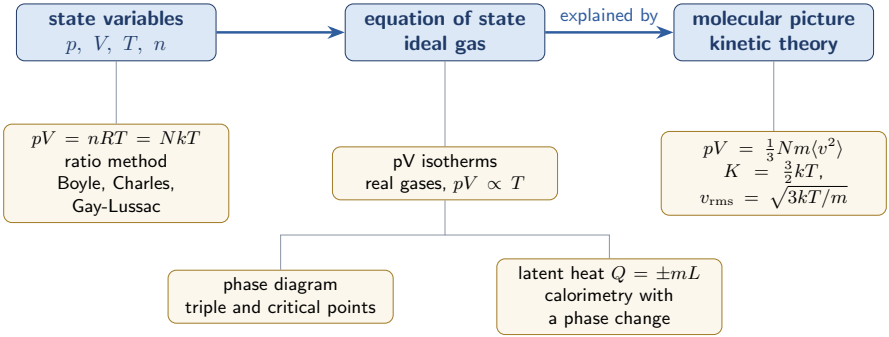
Weather and the atmosphere, diffusion, and why light gases escape from planets all follow from the speed distribution.

Physical meaning

The kinetic model connects the large-scale (p, V, T) to the molecular scale: the pattern for all of thermal physics.

But what if... the gas expands and pushes a piston outward while heat flows in? How much work does it do, and where does the energy go? → **Lecture 11**.

Lecture 10 concept map



MASTER PROCEDURE – THERMAL PROPERTIES

1. IDENTIFY The system, its phase and the process. Is the amount of gas fixed?
2. SKETCH The system, the path on the pV or phase diagram, or the molecule and wall.
3. UNITS Kelvin, absolute pressure, m^3, moles, M in kg/mol.
4. PREDICT The direction of change, a limiting case and the rough size.
5. SELECT THE MODEL Equation of state, ratio method, diagram, latent heat, calorimetry or kinetic theory.
6. EXECUTE Write the equations first, then substitute with units.
7. EVALUATE Units, signs, magnitude, limiting cases, and a second check.

SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–5
EXECUTE $\leftrightarrow$ step 6
EVALUATE $\leftrightarrow$ step 7

Physical meaning

Next lecture: the work done by a gas as its volume changes (the area under a path on the pV diagram) and the first law of thermodynamics.

THE SEVEN TOOLKITS

1	**State variables and units**	kelvin; absolute pressure; m^3; moles
2	**The ideal-gas equation**	$pV = nRT = NkT$; $n = m/M$
3	**The ratio method**	$p_1V_1/T_1 = p_2V_2/T_2$; what is constant?
4	**Reading pV and phase diagrams**	$T \propto pV$; paths; triple and critical points
5	**Heating curves and latent heat**	$mc\Delta T$ in a phase; $\pm mL$ at a change
6	**Calorimetry with a phase change**	decide the final state first; $\sum Q = 0$
7	**Kinetic theory**	$K = \frac{3}{2}kT$; $v_{\text{rms}} = \sqrt{3kT/m}$; scaling

The Master Procedure sits above these. Earlier toolkits (units, energy, signs, $Q = mc\Delta T$) are reused, not renamed.

Key ideas: the opening question answered

How do pressure, volume and temperature describe a gas, what happens at a change of phase, and what do molecules do?

- Use kelvin and absolute pressure in gas laws.
- A change of phase takes place at a definite temperature, which stays constant: $Q = \pm mL$ ($+$ if heat enters).
- Decide the final state before writing $\sum Q = 0$ with a phase change.
- Pressure comes from the collisions of molecules with the walls.
- The average translational kinetic energy per molecule, $\frac{3}{2}kT$, is proportional to the absolute temperature.
- At the same T molecules of different mass have the same average kinetic energy; the lighter ones are faster.

Where to look

Definitions: the vocabulary and equation-anatomy slides of each contact.

Common mistakes: the last slide of each contact.

Connections: the Connect boxes and slides.

Summary: core equations

Idea	Equation	Conditions and units
Amount	$m_{\text{total}} = nM, \quad N = nN_A$	M in kg/mol
Ideal gas	$pV = nRT = NkT$	kelvin, absolute p
Change of state	$p_1V_1/T_1 = p_2V_2/T_2$	fixed n
Heat, one phase	$Q = mc\Delta T$	J
Change of phase	$Q = \pm mL$	constant T
Kinetic pressure	$pV = \frac{1}{3}Nm(v^2)_{\text{av}}$	elastic wall collisions
Molecular energy	$K = \frac{3}{2}kT, \quad K_{\text{tr}} = \frac{3}{2}nRT$	translational
Rms speed	$v_{\text{rms}} = \sqrt{3kT/m} = \sqrt{3RT/M}$	m/s

- ☐ define the state variables and use n, N, M?
- ☐ convert to kelvin, absolute pressure and SI volume?
- ☐ use $pV = nRT$ and $pV = NkT$?
- ☐ use the ratio method and relate it to Boyle, Charles and Gay-Lussac?
- ☐ compare states on a pV diagram?
- ☐ read a phase diagram, including the triple and critical points?
- ☐ calculate $Q = \pm mL$ and a heating curve?
- ☐ solve calorimetry with a change of phase?
- ☐ state the assumptions of the kinetic model?
- ☐ explain pressure from molecular collisions?
- ☐ use $K = \frac{3}{2}kT$ and $v_{\text{rms}} = \sqrt{3kT/m}$?
- ☐ compare two gases at the same temperature?
- ☐ combine the gas law with the kinetic model?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra gas-law (including gauge pressure), pV, heating-curve, phase-change calorimetry, kinetic-theory and combined problems. Supplementary notes: the ideal gas as a model, the kinetic derivation, the Maxwell speed distribution, mean free path, equipartition (preview), real gases and the water phase diagram.

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